8

Imaging Gargantua

Black holes emit no light, so the only way to see Gargantua is by its influence on light from other objects. In *Interstellar* the other objects are an accretion disk (Chapter 9) and the galaxy in which it lives including nebulae and a rich field of stars. For the sake of simplicity, let's include only the stars for now.

Gargantua casts a black shadow on the field of stars and it also deflects the light rays from each star, distorting the stellar pattern that the camera sees. This distortion is the gravitational lensing discussed in Chapter 3.

Figure 8.1 shows a rapidly spinning black hole (let's call it Gargantua) in front of a field of stars, as it would appear to you if you were in Gargantua's equatorial plane. Gargantua's shadow is the totally black region. Immediately outside the shadow's edge is a very thin ring of starlight called the "ring of fire" that I intensified by hand to make the edge of the shadow more distinct. Outside that ring we see a dense